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Head Impact, Concussion, and Salary in the NFL: Is There a Compensating Wage-Risk Premium?

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Abstract

Sport epidemiological studies have estimated cumulative head impact and concussion rates by American football position group. Meanwhile, the sports economics literature has examined National football league (NFL) player-salary variation, and sports statisticians have studied NFL player productivity. Merging novel data and methodologies, we estimate whether players in position-groups with higher head impact or concussion risk are paid a compensating next-contract salary premium controlling for NFL experience, contract year, and productivity. For 2006 to 2017 NFL seasons, we consider all 1,162 fully observed player contracts and all 290 fully observed, non-rookie player contracts. Specifying two sets of contract-length frequency-weighted, contract-year (of signing) fixed effects linear regressions, we find robust evidence that players receive lagged, next-contract compensation for additional on-field productivity and experience. However, position groups with higher estimated cumulative head impact and, alternatively, higher estimated concussion rate are paid a significant and substantial next-contract *dispremium*. Results are consistent with an augmented compensating salary-differential theory, in which firm and employee share risk.

Keywords

NFL football, head impact, sports economics, concussion rate, compensating wage differential, labor economics

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Introduction

Occupational Hazards of NFL Play

NFL teams pay players for performance (Leeds & Kowalewski, 2001), as well as for position-specific skills (Roach, 2018). NFL players are exposed to repetitive head impacts. These hits are often delivered at high velocity and force, leading to risks of concussion and also sub-concussive forms of head trauma (Montenigro et al., 2017). There is a strong link between repetitive head trauma and development of neurodegenerative disease such as chronic traumatic encephalopathy (CTE), Parkinson's disease, dementia, and Alzheimer's disease (Montenigro et al., 2017; Owora et al., 2018). DeKosky et al. (2018) state, "The recent upsurge in interest in traumatic brain injury and concussion in sports, and in the study of neurodegenerative diseases in contact sports, was sparked by increasing evidence over the past 10 to 15 years that professional American football players, as well as professional athletes in most if not all contact sports, may develop short-term, career-threatening, and long-term brain diseases." Repetitive head impact contributes to neuroinflammation, which in turn contributes to pTau protein pathology (CTE) in several brain cortices (Cherry et al., 2016). Especially in stages 3 and 4, CTE presents as a visible, whole-brain disease in which pTau protein tangles accumulate (McKee et al., 2016). Alosco et al. (2017) find that a higher cumulative head impact index predicts significantly higher plasma Tau levels in former NFL players and that Tau might be a meaningful CTE biomarker.

At normal levels, pTau protein is important in assembling neuronal microtubules. When pTau becomes unregulated (e.g., in CTE cases), pTau levels spike, and the accumulation of pTau tangles interrupts "normal" neural activity. In turn, this presents elevated risk for CTE (McKee et al., 2009) and associated symptoms of short-term memory loss, impulsive behavior, depression, and difficulty planning and executing tasks (Asken et al., 2016). Moreover, elevated cumulative head impact risk represents an early mortality risk factor (Kmush et al., 2020), although this particular early mortality finding is not observed in the natural experimental setting of Venkataramani et al. (2018). Importantly, estimated head impact risk exposure varies by position in American football (Montenigro et al., 2017; Kmush et al., 2020). Therefore, head impact can be characterized as an occupational hazard in the NFL in terms of both long-term neurocognitive health risk exposure and mortality risk exposure. In addition to environment, risk of neurodegenerative pathology is also influenced by genetic variation (Harden & Koellinger, 2020), although individuals are largely uninformed as to their genetic predisposition.

In a related *Washington Post* article, Jenkins and Maese (2013) write, "The NFL contends it offers benefits—many extending beyond an athlete's employment—that are more generous than those offered in most professions. . .When it opposes [retired workers' compensation] claims, the NFL says it is seeking to limit what would otherwise be 'nearly unlimited exposure' for years-old injuries." Hence, the NFL argues that its compensation of active and recently retired players is remunerative of player injury such that it bears limited liability for the long-term health of retired players. There are existing counter-arguments for this contention. Byrd (2017) finds that a

large proportion of NFL contracts are not guaranteed against injury, for example. The present study seeks to examine whether head impact risk exposure of NFL players is revealed to be compensated within the observed NFL player salary distribution, as in the case of a standard compensating wage differential in labor economic theory.

Compensating Wage Differentials

The labor economics literature provides strong and consistent evidence that, in addition to productivity, workers across many industries are compensated for accepting additional early mortality risk exposure (see, e.g., Black & Kniesner, 2003; Lavetti, 2020). More generally, workers are compensated for accepting undesirable work conditions, whether these conditions impose additional early mortality risk or not (Guardado & Ziebarth, 2019). In sports labor settings, the literature typically finds evidence that players are compensated for salient forms of injury risk. Dole and Kassis (2010) find some evidence that NFL players at "scoring positions" are compensated more for playing home games on artificial turf, a substantially riskier surface for several types of leg injury than natural grass due to its unnatural "grab" and perhaps due to players having much more sport developmental experience on natural grass. Lavetti (2020) finds that Alaskan Bering Sea commercial fishermen (fisherwomen) are compensated a premium for undertaking riskier work conditions.

NFL teams face a hard cap of $224.8 million in 2023. An active NFL roster consists of 53 players, and there are 11 starters on offense and 11 distinct starters on defense. With 22 positions and 53 roster spots, teams can carry only 2 to 3 active players per position. As an anonymous referee on this manuscript suggests, this roster size pressure causes many teams to carry only two players at quarterback on game days, despite the importance of the position, while dressing three or more players at more versatile positions and positions for which backup players can serve on special teams. Despite injury protocols for players, there is therefore pressure to "play hurt" in some cases on game days, especially at positions like quarterback.

Despite this evidence, head trauma is less salient than many external injuries. The traditional diagnosis relies on induced behavioral symptoms (e.g., eyeball tracking) rather than something as concrete as an X-ray or MRI. As such, it is not clear that head injuries will command the standard wage premium observed in the literature for other injury classes.

The classic labor economic theory of compensating wage differentials is straightforward. We can consider two jobs, A and B, that are identical except that job B has a relatively undesirable aspect that is not observed in job A. For example, job A (B) might represent playing in the NFL at a position that features relatively low (high) head impact. Given this undesirable feature, labor supply for job B is expected to be shorter or weaker than labor supply for job A. That is, labor supply for job B is expected to be positioned to the left of labor supply for job A. There is case documentation of this effect in the NFL. Current and prospective NFL players have quit *due at least in part to concussion and/or long-term brain injury risk* (e.g., Chris Borland in 2015, Jack Miller in 2015, John Urschel in 2017, Joshua Perry in 2018, Luke Kuechly in

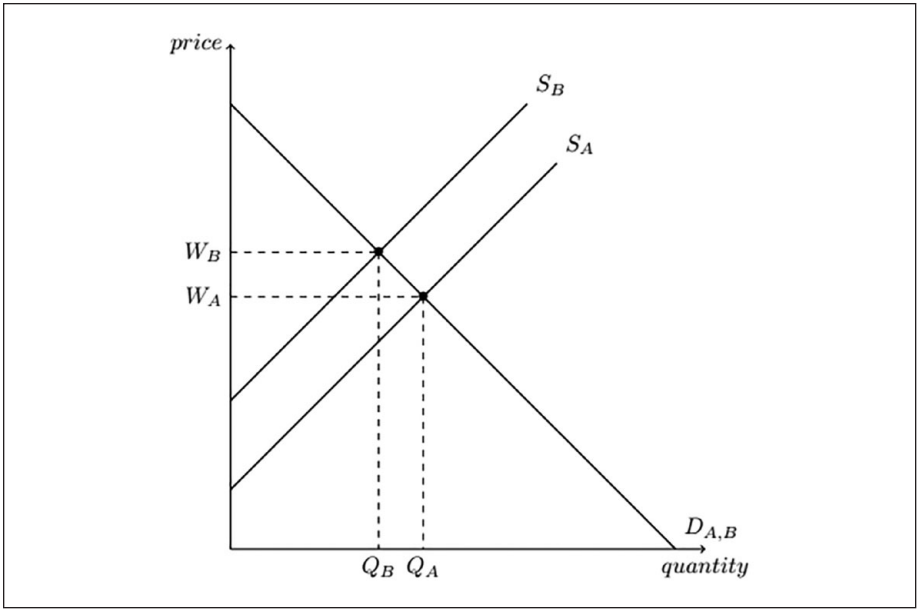

Figure 1. Effect of additional worker risk on labor market equilibrium.

2020, and Jordan Reed in 2021) according to their own attributions. For a given level of labor demand, then, we expect equilibrium salary for job B to be higher than equilibrium salary for job A. Figure 1 illustrates this comparison.

The traditional analysis states that an undesirable work condition is expected to decrease labor supply from S_A to S_B and increase the worker's equilibrium wage or salary from W_A to W_B. However, the case of head impact in professional sports may not be so straightforward. If teams believe that players at position B are more likely to miss games (e.g., due to concussion or other classes of head injury) than players at position A, then team *demand* for individual players at position B is also expected to be weaker than demand for individual players at position A, leading to the plot of Figure 2.

In this case, risk cuts in both directions. Players incur additional head impact risk. Team employers incur additional risk of lost work time (games played) for salaried workers. In turn, this decreases the expected output (wins) for the employer if the team does not have sufficient depth at high risk positions and increases the payroll costs given that replacement players must be contracted in more cases. In Figure 2, it is indeterminate in general whether the equilibrium salary of worker A or worker B is greater. Additional head impact risk from position A to position B is expected to decrease the labor supply as in a classical compensating wage differential setting. In the present case, however, head impact risk also has an effect on the likelihood that a player will miss a future game such that teams must budget, within the framework of a league-capped team payroll, for more backup players at positions with high head impact risk. Therefore, head impact risk in this setting is also expected to decrease the

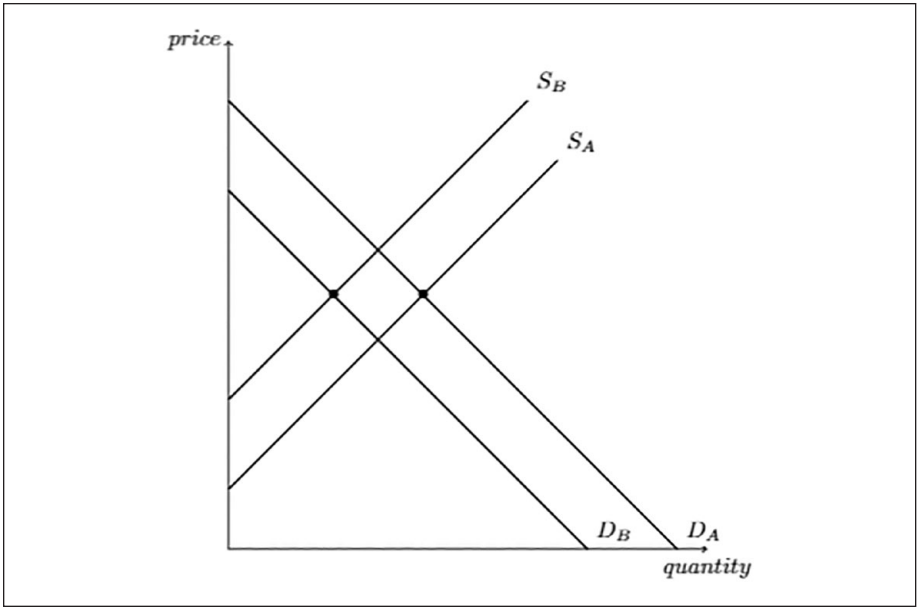

Figure 2. Effect of additional worker and team risk on labor market equilibrium.

demand for labor. In general, the direction of the change in equilibrium salary from worker A to worker B depends upon which shift dominates in magnitude.

This labor market analysis also suggests that concussive and sub-concussive head impact risk might be treated differently when NFL player valuations are assigned in the free agency market. The former type of risk represents an undesirable work condition that also results in a higher number of expected games missed. As such, concussive head impact risk is expected to constitute a shared risk setting, as represented in Figure 2. The latter type of risk represents an undesirable work condition that does not result in a higher number of expected games missed. Therefore, it is expected that only players will bear the risk of sub-concussive head impact, as represented in Figure 1. Given these differential effects, we expect sub-concussive head impact to raise observed, next-contract salaries and concussive head impact to have an indeterminate effect on observed, next-contract salaries in the NFL. Further, we know that all head impact is either concussive or sub-concussive. Therefore, the overall effect of head impact upon NFL salaries is a priori indeterminate according to the labor market analysis. The relationship(s) of interest will be obtained empirically herein and will not derive from any sort of determinate result from labor economic theory. Before we continue to the estimation, it is important to note that compensating wage premia are a market behavioral phenomenon rather than a legally stipulated phenomenon. Therefore, we are examining how decentralized, "invisible hand" forces related to head impact influence revealed, next-contract compensation in the NFL player labor market.

Data and Methods

Our empirical approach in this study is to estimate the effect of position-level expected head impact and, alternatively, position-level expected concussion rate upon a player's next-contract average annual salary, while controlling for a player's productivity level and years of NFL experience prior to the signing of the player's contract of observation. The regression models determining this relationship are further controlled in the following manners: (a) each player-contract level observation is frequency-weighted by contract length because team financial commitments and player returns increase linearly in number of contract years for a given average annual salary and (b) contract year (of signing) is specified to control for idiosyncratic variation in NFL player labor markets by calendar year. For example, there may be a shorter supply of talent from the NFL Draft in a year such that demand for veteran free agent players increases, *ceteris paribus.*

Player-season salary was obtained from Spotrac.com. To capture variation in player productivity, we use a new and proprietary variable known as Pro Football Focus Wins Above Average (WAA). This measure was obtained through an academic use agreement from the company Pro Football Focus. It was presented as a research finalist study at the 2020 MIT Sloan Sports Analytics Conference (Eager & Chahrouri, 2020) and is perhaps the first reliable, position-specific Wins Above Average measure for NFL players. As Pro Football Focus is a primary provider of player value data to NFL teams, WAA is more than a rough control for team estimates of player productivity within the study. Rather, NFL teams have available data on this exact variable when making veteran player contract decisions. As the data is proprietary, its use as a primary control in a player health study is novel to the best of our knowledge. To estimate position-level head impact risk, we utilize three different measures in alternative specifications of the regression model. These include position-clustered, season-level cumulative head impact index risk factor, as developed in Montenigro et al. (2017) and further established for NFL play in Kmush et al. (2020), and two distinct measures of position-clustered concussion rate (per 100 games played and, alternatively, per 1,000 Athletic Events), as obtained from Nathanson et al. (2016). Position clusters are specified in the extant literature as follows.

Quarterbacks, Receivers, Offensive Backs or Running Backs, Defensive Backs, Linebackers, Offensive Linemen, Defensive Linemen

Given that WAA is specified for offensive skill positions alone, only the first three position clusters are represented in our data. These include contracts to: *Quarterbacks* ($n=217$), *Receivers* ($n=631$), and *Offensive Backs* or *Running Backs* ($n=314$). It should be noted that the position cluster *Receivers* represents two listed positions, *Wide Receiver* and *Tight End*. In modern nomenclature, moreover, the *Offensive Backs* or *Running Backs* position cluster primarily represents *Running Backs* or *Full Backs*. Therefore, the analysis considers five playing positions that are clustered into three playing position clusters, as per the extant literature. Variation in position cluster will

be utilized to control for average position-cluster level concussion and head impact risk levels, as measured in the previous literature.

We take teams as having rational expectations of player productivity and player head impact risk factor such that the expectations of these values are taken into account when a player's next-contract salary is assigned. The empirical estimation will determine whether this assumption is valid. Our full data for the study includes all 1,162 NFL player contracts from 2006 to 2017 for which the specified variables were fully observed. This time window was selected because it allows for use of proprietary data for the variable Wins Above Average, which was obtained for these years via academic use agreement from Pro Football Focus. Another advantage of this time window is that it comprises a single collective bargaining agreement period such that the salary rules did not change during the study period. Sub-sample regressions examine all ($n = 290$) such NFL player contracts for which the player was not signing a rookie contract such that a past NFL player productivity estimate existed for the year prior to signing of observed contract (i.e., as an enhanced control for player's expected on-field productivity in next contract).

Conditional upon player productivity, experience, and contract year fixed effects—and frequency weighting observations by contract length—this study asks whether NFL players are compensated for additional head impact and concussion risk exposure. Given the traditional and augmented compensating risk differential theories developed, we construct the following hypothesis and empirical questions.

Hypothesis: As sub-concussive head impact risk typically does not lead to injury games/practices missed but is an undesirable work condition, traditional compensating risk differential theory (Figure 1) applies. The partial effect of sub-concussive head impact risk is to increase NFL salaries, *ceteris paribus*.

Empirical Question 1: As concussive head impact risk typically leads to injury games/practices missed and is an undesirable work condition, the augmented compensating risk differential theory for two-way risk (Figure 2) applies. The sign of the partial effect of concussive head impact risk is therefore *a priori* indeterminate (an empirical question).

Empirical Question 2: Given that all head impact is either concussive or sub-concussive, the overall effect of head impact risk is *a priori* indeterminate (an empirical question).

NFL teams observe a recent, NFL-level productivity estimate prior to signing veteran free agent players. Therefore, we refer to the non-rookie contract group as the "ideal sub-sample" for the analysis. We consider the research hypothesis and questions first for all observed player contracts from 2006 to 2017. To account for player salary constraints and lack of prior NFL productivity data for players in rookie contract, we also consider the research hypothesis and questions for all 2006 to 2017 NFL player contracts in which the player was beyond his rookie contract. Given that all these sub-sampled players earn free agency salaries that are based on prior NFL productivity, sub-sample regressions will provide the most rigorous test of our research hypothesis/questions. Thus, we summarize key variables *for the ideal sub-sample* in the next section.

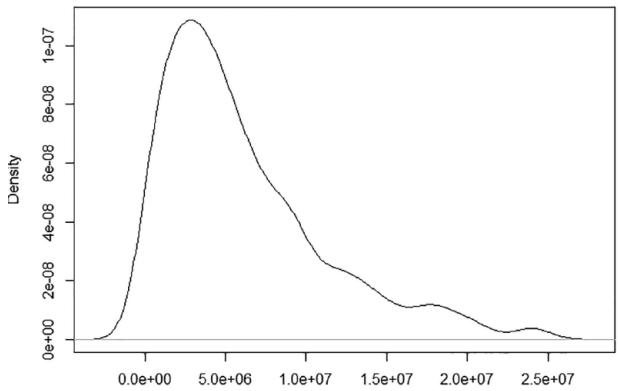

Figure 3. Density plot of salary for NFL players beyond rookie contract, 2006 to 2017.

Data Summary and Visualization

We now summarize key variables for the ideal sub-sample. Figure 3 provides a density plot of the dependent variable, player-season average annual *salary* in contract of observation.

The salary of players not on a rookie contract is approximately normally distributed with a mean of $5.69 million, a standard deviation of $5.19 million, and some degree of right skewness (+1.33). Despite this skewness, standard errors are asymptotically efficient given the large sample size such that linear regression inferential statistics are valid in this case even without transforming the dependent variable. We rely on this asymptotic efficiency to avoid a transformation of y that would diminish the interpretability of the model. We also obtain a density plot for WAA in Figure 4.

The mean of WAA is 0.11, and the standard deviation is 0.39 for sub-sampled players not on a rookie contract. The WAA variable measures player win value above (below) leaguewide average at playing position. When weighted by number of snaps for each player in a season, then, average WAA *at any position* during a season must equal 0. The occurrence that unweighted mean for WAA is slightly positive for players not on a rookie contract indicates that this sub-sample of players almost certainly performed slightly better *in the sample* than those players on a rookie contract. We expect the *snap-weighted* mean for this sub-sample to be even more positive, as there is an expected positive correlation between player win value and player snap count deriving from a team's objective to sort the roster depth chart to achieve more wins and not fewer. It stands to reason that players not on a rookie contract appear to be more win valuable, on average, than players on a rookie contract, as players are retained or not retained on the basis of their first contract performance.

We also summarize the following variables: years of NFL experience prior to contract of observation, Concussions_Per_100_GPs (Expected Concussions per 100 Games Played by Position Cluster), and Head_Impact_Risk_Factor by position-cluster

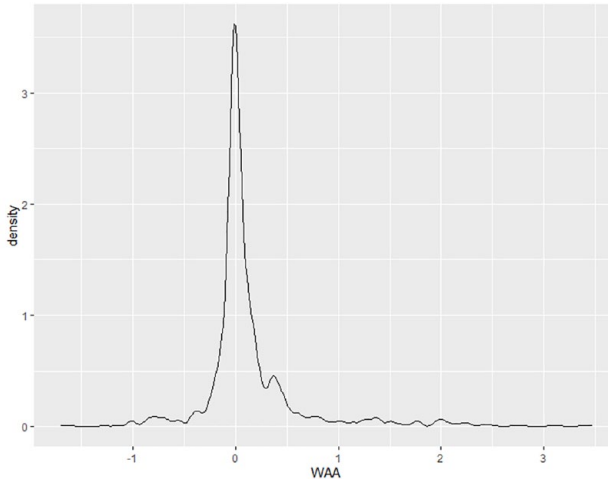

Figure 4. Wins above average for NFL players not on rookie contract, 2006 to 2017.

(as developed, validated, and specified in cumulative head impact studies; see, e.g., Montenigro et al., 2017 or Kmush et al., 2020; head impact position-cluster risk data obtained from Kmush et al., 2020). In the sub-sample that does not include first-contract players, mean prior NFL experience level is 6.28 years, and standard deviation of experience is 3.61 years. Pro-rated to 20 games played, which represents roughly a full NFL season when including pre-season and post-season, NFL players in the sample averaged 0.49 reported concussions ($SD = 0.14$) and a head impact risk factor of 329.0 units ($SD = 95.6$) according to the variables Concussions_Per_100_GPs and Head_Impact_Risk_Factor.

In the regression section, we also utilize a concussion risk variable that takes into account practices, Concussions_Per_1000_AEs (Concussions per 1,000 athletic events, where athletic events are games or practices; data for both concussion rate measures obtained from Nathanson et al., 2016). As certain positions such as *Quarterback* may receive special protection during practice compared to during games, this alternative measure of concussion risk is also important. The mean Concussions_Per_1000_AEs in the sub-sample is 8.80 concussions with a standard deviation of 1.68 concussions. According to practice rules, as set forth in the NFL Collective Bargaining Agreement, teams hold approximately 75 athletic events per season. Then, the average number of concussions per 75 athletic events, or season worth of athletic events, is $8.80.(75/1000)$ or 0.66 concussions, with the majority of that share coming from game play. For a player engaging in 75 athletic events, with 20 of them being games, sample estimates suggest that approximately 0.74 proportion of concussions will occur during game play. On the other hand, game play accounts for only 0.27 proportion of those 75 athletic events. In other words, this sample estimate suggests that about three times as many concussions would arise from games during

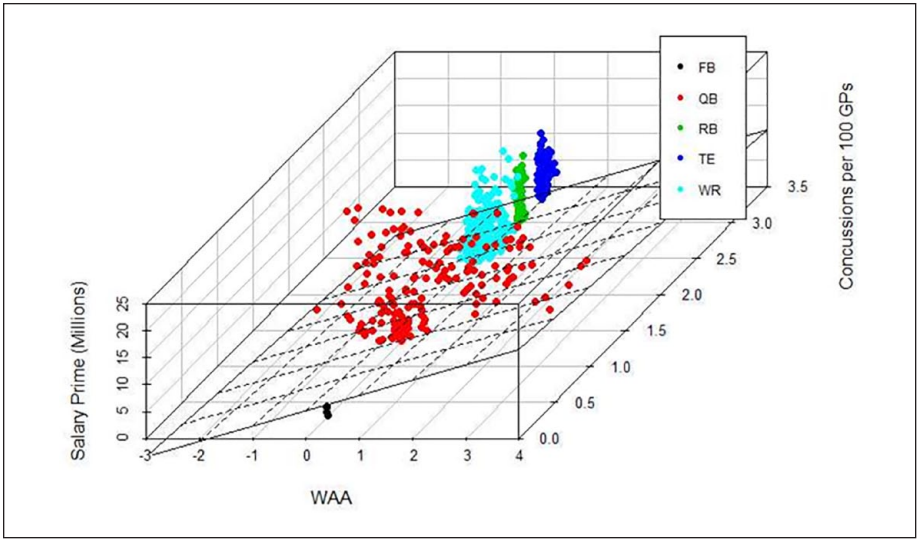

Figure 5. Salary, WAA, concussion rate for NFL players not on rookie contract, 2006 to 2017.

this 75-athletic event exposure but that practices would occur about three times as often as games. A rough sample estimate from these measures, then, would suggest that a game exposure carries about nine times the concussion risk as a practice exposure. In terms of concussion risk exposure, then, these sample estimates suggest that the NFL's recent addition of a 17th regular season game is akin to beginning preseason practices a week and a half earlier from the perspective of estimated head impact.

We now generate a three-dimensional scatter plot with trend plane to visualize the baseline relationship between Salary, WAA, and Concussions_Per_100_GPs.

Whereas Salary is highly responsive to WAA along the (*x,z*) Cartesian plane, the relationship between Salary and concussion rate along the (*y,z*) Cartesian plane is much weaker. As concussion rate rises, moreover, we observe that Salary actually decreases in the sample. Of course, the three-dimensional scatter plot and trend plane representation of Figure 5 does not include a full set of controls by definition. In the regression models to follow, we examine whether these apparent relationships hold conditional upon such a full set of controls. These models differ from standard fixed effects linear models in that the optimization function weights observations by contract length of player *i* playing at position cluster *j* at time *t*, $c_{i,j,t}$, when minimizing the sum of squared errors. That is, coefficient estimates, b_j, represent $argmin\ \Sigma_{\forall i,k}\, c_{i,j,t}.\left(y_{i,j,t} - \bar{y}_{i,j,t}\right)^2$ rather than $argmin\ \Sigma_{\forall i,k}\left(y_{i,j,t} - \bar{y}_{i,j,t}\right)^2$. Models (1–6) represent whole sample regressions. Models (7–12) represent ideal sub-sample regressions, where Model 7 (8, 9, 10, 11, 12) has the same specification as Model 1 (2, 3, 4, 5, 6) with the exception that WAA is lagged to the "contract year"—the season before signing of contract—in Models 7–12 (i.e., for the sub-sample in which lagged productivity is available).

$$\begin{aligned}\text{Salary}_{i,j,t} &= b_0 + b_1\text{WAA}_{i,j,t} + b_2\text{experience}_{i,j,t} \\ &+ b_3\text{Head_Impact_Risk_Factor}_{j,t} \\ &+ \gamma.\ \text{Year_Fixed_Effects}_t + e_{i,j,t}\end{aligned} \tag{1}$$

$$\begin{aligned}\text{Salary}_{i,j,t} &= b_0 + b_1\text{WAA}_{i,j,t} + b_2\text{experience}_{i,j,t} + b_3\text{experience}^2{}_{i,j,t} \\ &+ b_4\text{Head_Impact_Risk_Factor}_{j,t} \\ &+ \gamma.\text{Year_Fixed_Effects}_t + e_{i,j,t}\end{aligned} \tag{2}$$

$$\begin{aligned}\text{Salary}_{i,j,t} &= b_0 + b_1\text{WAA}_{i,j,t} + b_2\text{experience}_{i,j,t} \\ &+ b_3\text{Concussions_Per_1000_AEs}_{j,t} \\ &+ \gamma.\text{Year_Fixed_Effects}_t + e_{i,j,t}\end{aligned} \tag{3}$$

$$\begin{aligned}\text{Salary}_{i,j,t} &= b_0 + b_1\text{WAA}_{i,j,t} + b_2\text{experience}_{i,j,t} + b_3\text{experience}^2{}_{i,j,t} \\ &+ b_4\text{Concussions_Per_1000_AEs}_{j,t} + \gamma.\text{Year_FEs}_t + e_{i,j,t}\end{aligned} \tag{4}$$

$$\begin{aligned}\text{Salary}_{i,j,t} &= b_0 + b_1\text{WAA}_{i,j,t} + b_2\text{experience}_{i,j,t} \\ &+ b_3\text{Concussions_Per_1000_GPs}_{j,t} \\ &+ \gamma.\text{Year_Fixed_Effects}_t + e_{i,j,t}\end{aligned} \tag{5}$$

$$\begin{aligned}\text{Salary}_{i,j,t} &= b_0 + b_1\text{WAA}_{i,j,t} + b_2\text{experience}_{i,j,t} + b_3\text{experience}^2{}_{i,j,t} \\ &+ b_4\text{Concussions_Per_1000_GPs}_{j,t} + \gamma.\text{Year_FEs}_t + e_{i,j,t}\end{aligned} \tag{6}$$

$$\begin{aligned}\text{Salary}_{i,j,t} &= b_0 + b_1\text{WAA}_{i,j,t-1} + b_2\text{experience}_{i,j,t} \\ &+ b_3\text{Head_Impact_Risk_Factor}_{j,t} \\ &+ \gamma.\text{Year_Fixed_Effects}_t + e_{i,j,t}\end{aligned} \tag{7}$$

$$\begin{aligned}\text{Salary}_{i,j,t} &= b_0 + b_1\text{WAA}_{i,j,t-1} + b_2\text{experience}_{i,j,t} \\ &+ b_3\text{experience}^2{}_{i,j,t} + b_4\text{Head_Impact_Risk_Factor}_{j,t} \\ &+ \gamma.\text{Year_FEs}_t + e_{i,j,t}\end{aligned} \tag{8}$$

$$\begin{aligned}\text{Salary}_{i,j,t} &= b_0 + b_1\text{WAA}_{i,j,t-1} + b_2\text{experience}_{i,j,t} \\ &+ b_3\text{Concussions_Per_1000_AEs}_{j,t} \\ &+ \gamma.\text{Year_Fixed_Effects}_t + e_{i,j,t}\end{aligned} \tag{9}$$

$$\begin{aligned}\text{Salary}_{i,j,t} &= b_0 + b_1\text{WAA}_{i,j,t-1} + b_2\text{experience}_{i,j,t} \\ &+ b_3 exp^2{}_{i,j,t} + b_4\text{Concussions_Per_1000_AEs}_{j,t} \\ &+ \gamma.\text{Year_Fixed_Effects}_t + e_{i,j,t}\end{aligned} \tag{10}$$

$$\begin{aligned}\text{Salary}_{i,j,t} &= b_0 + b_1\text{WAA}_{i,j,t-1} + b_2\text{experience}_{i,j,t} \\ &+ b_3\text{Concussions_Per_1000_GPs}_{j,t} \\ &+ \gamma.\text{Year_Fixed_Effects}_t + e_{i,j,t}\end{aligned} \tag{11}$$

$$\begin{aligned}\text{Salary}_{i,j,t} &= b_0 + b_1\text{WAA}_{i,j,t-1} + b_2\text{experience}_{i,j,t} \\ &+ b_3 exp^2{}_{i,j,t} + b_4\text{Concussions_Per_1000_GPs}_{j,t} \\ &+ \gamma.\text{Year_Fixed_Effects}_t + e_{i,j,t}\end{aligned} \tag{12}$$

In the first six models, Salary for player *i* playing at position *j* in year *t* is modeled as a linear or polynomial function of game productivity of player *i* at position *j* in year *t*, experience of player *i* at position *j* in year *t*, experience squared of player *i* at position *j* in year *t*, concussion or general head impact risk for position *j* in year *t* (given the player's listed position cluster), and year fixed effects. In the last six models, specifications are identical with the exception of the productivity lag. In the empirical setting of the final six regressions, teams have lagged contract-year productivity values, $\text{WAA}_{i,j,t-1}$, that were observed at time $t-1$, or in the season before the contract of observation was signed. As Pro Football Focus is a major provider of player value data to NFL teams, $\text{WAA}_{i,j,t-1}$ is not simply a cursory control for player productivity. Rather, the specific $\text{WAA}_{i,j,t-1}$ measure was available to teams when making veteran player contract decisions. At the time of contract signing, $\text{WAA}_{i,j,t-1}$ represents a player's most recent seasonal productivity value from NFL play. Teams can therefore use this information to set expectations for future productivity. In the first six regressions, we can only control for a player's revealed, post-contract productivity and assume that teams held rational expectations over player outcomes when signing players to contracts. Given this informational distinction, we will focus more closely on the refined results of Models 7 to 12.

Results and Discussion

Results of the 12 models are presented in Tables 1 and 2.

Despite the refined specifications of Models 7 to 12—given their use of lagged player productivity—both sets of regressions provide very similar results in terms of qualitative and significance results. These similarities support the notion that teams have rational expectations on player productivity in the absence of prior NFL player data (e.g., from scouting evaluation, college performance, and biometrics). The two sets of models are also fairly similar in terms of overall explanatory power. *R*-squared

Table 1. Regression output for models 1 to 6.

	Dependent variable					
	Salary					
	(1)	(2)	(3)	(4)	(5)	(6)
WAA	3,033,171.000***	3,112,547.000***	3,005,377.000***	3,082,936.000***	2,917,938.000***	2,994,588.000***
	(275,646.300)	(272,581.300)	(275,185.400)	(272,013.900)	(267,394.900)	(263,875.800)
Experience	668,616.500***	984,753.400***	671,730.300***	991,563.900***	632,049.900***	958,060.600***
	(26,761.810)	(63,056.970)	(26,663.660)	(62,976.210)	(26,280.550)	(60,912.970)
Experiencesq		−28,185.720***		−28,485.580***		−29,138.790***
		(5,104.387)		(5,095.469)		(4,927.583)
Head_Impact_Risk_Factor	−5,361.808***	−5,880.674***				
	(906.559)	(900.150)				
Concussions_Per_1000_AEs			−317,928.300***	−348,226.200***		
			(50,791.180)	(50,432.510)		
Concussions_Per_100_GPs					−1,267,566.000***	−1,324,107.000***
					(119,950.200)	(118,614.700)
Constant	2,618,325.000***	2,366,931.000***	3,657,183.000***	3,497,600.000***	4,170,413.000***	3,879,713.000***
	(328,757.800)	(327,826.500)	(470,408.300)	(465,258.100)	(339,087.900)	(337,817.400)
Observations	1,162	1,162	1,162	1,162	1,162	1,162
R^2	0.470	0.483	0.471	0.485	0.502	0.516
Adjusted R^2	0.468	0.481	0.470	0.484	0.500	0.515
Residual Std. Error	5,201,102.000	5,136,112.000	5,191,969.000	5,125,450.000	5,041,579.000	4,969,223.000
	(*df* = 1158)	(*df* = 1157)	(*df* = 1158)	(*df* = 1157)	(*df* = 1158)	(*df* = 1157)
F Statistic	341.774***	270.481***	344.336***	272.812***	388.558***	308.709***
	(*df* = 3; 1158)	(*df* = 4; 1157)	(*df* = 3; 1158)	(*df* = 4; 1157)	(*df* = 3; 1158)	(*df* = 4; 1157)

Note. *$p < .1$; **$p < .05$; ***$p < .01$.

Table 2. Regression output for models 7 to 12.

	Dependent variable					
	Salary					
	(7)	(8)	(9)	(10)	(11)	(12)
WAA_contract_year	5,689,254.000***	5,833,761.000***	5,594,625.000***	5,728,524.000***	5,099,163.000***	5,216,688.000***
	(647,947.900)	(649,884.400)	(644,334.400)	(646,147.700)	(621,390.000)	(623,688.600)
Experience	479,454.100***	828,767.400***	476,640.700***	812,550.200***	363,741.500***	656,609.600***
	(75,155.450)	(202,615.400)	(74,694.840)	(201,181.700)	(74,224.870)	(193,251.700)
Experiencesq		−25,044.410*		−24,047.770*		−20,924.810
		(13,499.540)		(13,381.380)		(12,754.860)
Head_Impact_Risk_Factor	−7,197.370***	−7,718.188***				
	(2,476.504)	(2,481.927)				
Concussions_Per_1000_AEs			−492,283.000***	−512,327.100***		
			(142,878.700)	(142,761.400)		
Concussions_Per_100_GPs					−2,137,012.000***	−2,141,249.000***
					(335,879.000)	(334,900.200)
Constant	4,720,660.000***	3,945,299.000***	6,749,496.000***	6,015,655.000***	8,433,453.000***	7,646,745.000***
	(1,006,639.000)	(1,086,010.000)	(1,413,550.000)	(1,466,088.000)	(1,095,368.000)	(1,192,787.000)
Observations	290	290	290	290	290	290
R^2	0.332	0.340	0.340	0.347	0.398	0.403
Adjusted R^2	0.325	0.331	0.333	0.338	0.391	0.395
Residual Std. Error	6,445,747.000	6,418,406.000	6,408,585.000	6,383,750.000	6,121,356.000	6,103,335.000
	(df = 286)	(df = 285)	(df = 286)	(df = 285)	(df = 286)	(df = 285)
F Statistic	47.419***	36.728***	49.079***	37.904***	62.950***	48.164***
	(df = 3; 286)	(df = 4; 285)	(df = 3; 286)	(df = 4; 285)	(df = 3; 286)	(df = 4; 285)

Note. *$p < .1$; **$p < .05$; ***$p < .01$.

values fall between 0.33 and 0.40 for the latter model set, a typical ballpark range for labor econometric studies.

We will further interpret the results of Models 7 to 12. The coefficient for contract year productivity is positive and highly significant. Each additional unit of WAA_contract_year increases the player's next contract annualized salary by between $5.1 million and $5.8 million, or a little under the mean salary for this sub-sample. As a unit change in WAA represents a drastic change in value at the player-season level, it is not surprising that the marginal salary effect of a unit gain in WAA is almost equal to mean salary. In the non-rookie contract sub-sample, a +0 WAA player was at the 40th percentile of WAA value, whereas a +1 WAA player was at the 97th percentile. Moreover, this result is in line with "cost of win" estimates in other professional sports (see, e.g., Swartz, 2017 for an analysis of the cost of a win through free agent acquisitions in MLB). The coefficients for Experience and Experience squared provide significant evidence that salary has an inverted-U relationship with NFL experience, as consistent with professional athlete aging curves (see, e.g., Ghimire et al., 2020) and career salary profiles in many fields (see, e.g., Luong & Hébert, 2009).

Conditional upon the controls discussed above, we find robust evidence that teams pay a significant *dispremium* to players with a higher concussion risk *and* overall head impact risk. We find this result across the 12 model specifications and also across three distinct measures of head impact risk: Head_Impact_Risk_Factor, Concussions_Per_100_GPs, and Concussions_Per_1000_AEs. We can prorate the second variable to per 20 games played, where this is the typical number of preseason, regular season, and postseason games combined in an NFL season. If one's expected concussions per 20 games increases by a unit, this is equivalent to one's expected concussions per 100 games increasing by 5 units. Therefore, the marginal coefficient prorated to 20 games will have 5 times the strength. As this is a rate variable, effect magnitude is the reciprocal of exposure. The sub-sample mean concussion rate per season in the sub-sample is 0.49. Therefore, at mean concussion rate and salary, a player loses an estimated [0.49 (100/20) $2.1412 million] or $5.25 million to decreased team demand for individual players deriving from accumulated head impact risk/injury (according to Model 12). One can think of this result as follows: The average sampled player earned $5.69 million per season but would have earned an estimated $10.94 million per season if he were assured never to be injured due to head impact. While this appears to be a large effect, we observe that (a) it will be mitigated by the effect of sub-concussive head impacts and (b) teams are not only responding to missed practices and games when considering high concussion risk positions. Teams are also responding to what the *Washington Post* story of Jenkins and Maese (2013) describes as "nearly unlimited exposure" to post-career player medical liability. If teams expect that they will bear some of the postretirement healthcare costs of players, especially for those at high concussion rate positions, they may be retaining some of that salary in up-front salary negotiations as a player demand effect.

The results also provide novel estimates as to how teams discount player salaries based on head impact more generally. The mean head impact risk factor in the model is 329.0 units. Therefore, at mean head impact risk and salary, a player loses an

estimated 7,718.19, 329.0 or $2.54 million in annual salary to decreased team demand for individual players deriving from accumulated head impact risk/injury (according to Model 8). This is the net effect of head impact upon salary: a $2.54 million *dispremium* to the player. The two possible sources of head impact are concussive head impact and sub-concussive head impact. One can therefore think of the net effect of head impact upon salary as follows: The average sampled player earned $5.69 million per season, loses $5.25 million in concussion risk dispremium, and then gains back ($5.25 million–$2.54 million) $2.71 million in sub-concussive head impact risk premium. On net, the player would have earned an estimated $8.06 million per season if he bore no head impact risk, *ceteris paribus*.

This suggests that, while players are paid a *dispremium* for concussive head impacts (given missed games and practices), they are paid a *premium* for sub-concussive head impacts (given no expected missed games or practices). One can think of (part of) the $2.54 million net dispremium as being set aside to sign more depth at that player's position, given the effects of head injury. One can further think of teams as retaining part of the $2.54 million net dispremium against future additional player healthcare liabilities, where this retention is due to a net player demand decrease given the likelihood of future liability. Of course, the team may set aside additional funds given other risks, such as other types of injury. While these differential effects are interesting and important in their own right, the net result of the study is that, on net, NFL teams are not paying a premium but rather a dispremium for player head impact exposure.

Conclusion

In a *Washington Post* article related to the present topic, Jenkins and Maese (2013) write, "The NFL contends it offers benefits—many extending beyond an athlete's employment—that are more generous than those offered in most professions. . . When it opposes [retired workers' compensation] claims, the NFL says it is seeking to limit what would otherwise be 'nearly unlimited exposure' for years-old injuries." Hence, the NFL argues that its compensation of active and recently retired players is remunerative of player injury such that it bears limited liability for the long-term health of retired players. The present study examines whether head impact risk exposure of NFL players is compensated within the NFL player next-contract salary distribution, as in the case of a standard compensating wage differential within labor economic theory. While players are compensated for on-field productivity and according to NFL experience, we find that the labor demand effect of head impact in the NFL dominates the labor supply effect. We find strong statistical evidence that NFL players are not compensated for, but rather receive a substantial net next-contract salary decrease from, additional head impact risk exposure.

The estimated dispremium for overall head impact is *less* than that for concussions, although concussive head impact is a subset of overall head impact. This suggests that, while players are paid a dispremium for concussive head impacts (given missed games and practices), they are paid a (smaller) *premium* for sub-concussive head impacts

(given no expected missed games or practices). Traditional compensating salary differential theory explains the former effect, and the latter effect is consistent with an augmented compensating salary differential theory, in which employing firm and employee share risk.

While these differential effects are interesting and important in their own right, the net result of the study is that NFL teams are not paying a premium but rather an estimated $2.54 million *dispremium* per player season, estimated at sample means, for head impact risk exposure. This result owes to the shared-risk setting of the study. Players and teams bear risk for player concussive head impact, where it is found empirically herein that team risks dominate in terms of salary effect. One can think of (part of) the $2.54 million net dispremium as being set aside to sign more depth at that player's position, given the effects of head injury. One can further think of teams as retaining part of the $2.54 million net dispremium against future additional player healthcare liabilities, where this retention is due to a net player demand decrease given the likelihood of future liability.

The present study, and subsequent studies along the same path, has potentially important implications for NFL team long-term healthcare liabilities. Rather than paying for head impact exposure, it appears that teams are allocating salaries so as to retain dollars against future projected healthcare liabilities. According to the results herein, there is no empirical evidence that teams have already allocated dollars to compensate players for head impact incurred during NFL play. On the contrary, our results suggest that teams pay large premia for win productivity and NFL experience but claw back some of those dollars in response to additional head impact exposure. That is, the results suggest that teams, on net, have allocated salaries so as to cover their own downstream risks associated with player head impact.

As an anonymous referee for the manuscript suggests, future work can examine the link between salary and head impact in the presence of improved productivity data of the analytic era. In a model that does not focus on risk level but rather position, this link can also be examined by position group.

Declaration of Conflicting Interests

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